Year 11 Maths Advanced

Functions & Graphs, Trigonometry Set 1

DA TUITION
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Student Name: ______________________________ **Class:** ____________

Time Allowed: 201 minutes **Total Marks:** 134

Section 1

1. Find the domain and range of the function $f(x) = \sqrt{4 - x^2}$. **(2 marks)**

2. Solve the equation $|2x - 3| = 7$. **(2 marks)**

3. Given $f(x) = x^2 - 3x$, evaluate $f(-2)$. **(1 mark)**

4. State the domain and range of the function $y = \frac{1}{x-3} + 2$. **(2 marks)**

5. Solve the inequality $3 - 2x < 7$. **(1 mark)**

6. Let $f(x) = x^2$ and $g(x) = x + 3$. Find the expression for the composite function $f(g(x))$. **(1 mark)**

7. Show whether the function $f(x) = x^3 - 3x$ is even, odd, or neither. **(2 marks)**

8. Solve the inequality $|x + 4| \leq 3$. Write your answer in interval notation. **(2 marks)**

9. Sketch the graph of $y = |x - 2|$, labelling the vertex and all intercepts. **(2 marks)**

10. Find the domain of the function $f(x) = \frac{1}{\sqrt{x^2 - 9}}$. **(2 marks)**

11. Solve the inequality $|2x - 1| > 5$ and sketch the solution on a real number line. **(2 marks)**

12. If $f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 1}$, find the domain of the composite function $g(f(x))$. **(2 marks)**

13. Sketch the parabola $y = (x - 1)^2 - 4$, clearly labelling the vertex, axis of symmetry, and all intercepts. **(2 marks)**

14. Solve the algebraic inequality $\frac{x+2}{x-3} \leq 0$. **(2 marks)**

15. A function is defined by $f(x) = \frac{x}{x-1}$ for $x \neq 1$. Show that $f(f(x)) = x$. **(2 marks)**

16. Determine the range of the function $y = 3 - |x+1|$. **(2 marks)**

17. Sketch the graph of the piecewise function:

$$f(x) = \begin{cases} -x-1, & \text{if } x < 0 \\ x^2 - 1, & \text{if } x \geq 0 \end{cases}$$

(2 marks)

18. The graph of $y = f(x)$ is reflected in the y-axis and then translated 3 units vertically upwards. Write down the equation of the resulting graph in terms of $f(x)$. **(2 marks)**

19. Solve the equation $x^2 - 4 = |x+2|$. **(2 marks)**

20. Find the domain and range of the function $y = -\sqrt{2x+6}$. **(2 marks)**

21. Solve the inequality $x^2 - 5x + 6 > |x-3|$. **(3 marks)**

22. Let $f(x) = \frac{1}{x}$ and $g(x) = 1 - x^2$. Determine the domain and range of the composite function $(f \circ g)(x)$. **(3 marks)**

23. Sketch the region in the Cartesian plane defined by the inequalities $y \geq x^2 - 4$ and $y \leq |x|$ simultaneously. **(3 marks)**

24. Find all values of the constant k for which the quadratic equation $x^2 - (k-2)x + 9 = 0$ has no real roots. **(3 marks)**

25. Find the rule, domain, and range for the inverse function $f^{-1}(x)$, given $f(x) = \frac{2x+1}{x-3}$. **(3 marks)**

26. Solve the inequality $\frac{|x-2|}{x^2-4} \geq 1$. **(3 marks)**

27. Consider the functions $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4x + 3$. Find the exact domain of the composite function $(f \circ g)(x)$. **(3 marks)**

28. A function $f(x)$ satisfies the functional equation $f(x) + 2f(1-x) = x^2$ for all real x. Determine the explicit algebraic expression for $f(x)$. **(3 marks)**

29. Find all values of the gradient m for which the line $y = mx - 5$ does not intersect the parabola $y = x^2 - 2x - 1$. **(3 marks)**

30. Sketch the graph of $y = \frac{|x^2-1|}{x-1}$. Clearly label any intercepts, asymptotes, or holes (points of discontinuity). **(3 marks)**

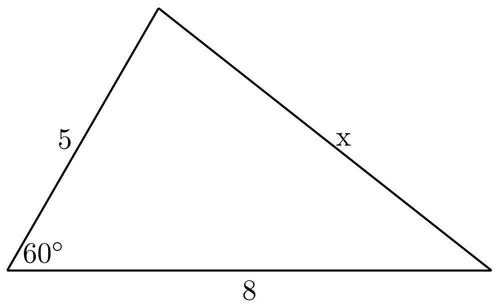
Section 2

1. Convert the angle 225° to radians, expressing your answer in terms of π. **(1 mark)**

2. Find the exact value of $\sin\left(\frac{5\pi}{3}\right)$. **(1 mark)**

3. Solve the trigonometric equation $\cos\theta = -\frac{1}{2}$ for $0 \leq \theta \leq 2\pi$. **(2 marks)**

4. Find the length of the unknown side x in the triangle shown below:



(2 marks)

5. State the amplitude and period of the function $y = 3\sin(2x)$. **(2 marks)**

6. Simplify the trigonometric expression $\frac{\sin(\pi-\theta)}{\sin(\frac{\pi}{2}-\theta)}$. **(2 marks)**

7. In $\triangle ABC$, side $a = 7$ cm, side $b = 10$ cm, and $\angle C = 45°$. Find the exact area of the triangle. **(2 marks)**

8. Given that $\tan\theta = -\frac{3}{4}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\cos\theta$. **(2 marks)**

9. Solve $2\sin\theta - 1 = 0$ for $0° \leq \theta \leq 360°$. **(2 marks)**

10. Use complementary angle results to simplify the expression $\cos(90° - \theta)\tan(\theta)$. **(1 mark)**

11. Solve the equation $\tan^2\theta - 3 = 0$ for $0 \leq \theta \leq 2\pi$. **(2 marks)**

12. Prove the identity $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = 2\sec^2\theta$. **(2 marks)**

13. Find the exact value of $\cos\left(-\frac{7\pi}{6}\right)$. **(2 marks)**

14. In $\triangle ABC$, $a = 8$ cm, $b = 12$ cm, and $\angle A = 30°$. Find all possible values of $\angle B$ to the nearest degree. **(2 marks)**

15. Solve the equation $2\cos^2\theta + \sin\theta - 1 = 0$ for $0° \leq \theta \leq 360°$. **(2 marks)**

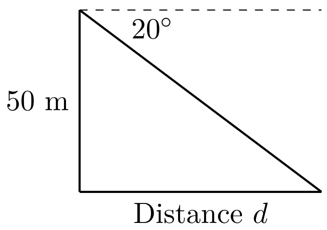
16. Sketch the graph of $y = 2\cos(x) + 1$ for $0 \leq x \leq 2\pi$. **(2 marks)**

17. A sector of a circle of radius 6 cm has an arc length of 4π cm. Find the exact area of the sector in terms of π. **(2 marks)**

18. Solve the equation $\sin\left(2\theta - \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}$ for $0 \leq \theta \leq \pi$. **(2 marks)**

19. Express the trigonometric expression $\cos\theta + \sin\theta\tan\theta$ as a single trigonometric ratio. **(2 marks)**

20. An observer standing on top of a vertical cliff of height 50 m looks down at a boat on the sea. The angle of depression is 20°. Find the horizontal distance of the boat from the base of the cliff, rounded to 1 decimal place.



(2 marks)

21. Solve the equation $2\sin^2\theta - 5\sin\theta\cos\theta + 2\cos^2\theta = 0$ for $0 \leq \theta \leq \pi$. **(3 marks)**

22. Prove the trigonometric identity: $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\csc\theta$. **(3 marks)**

23. Sketch the graph of $y = 3\sin\left(2x - \frac{\pi}{2}\right)$ for $0 \leq x \leq \pi$, labelling all intercepts, the amplitude, and the phase shift. **(3 marks)**

24. In $\triangle ABC$, the side lengths are $a = x+1$, $b = x$, and $c = x-1$. If the largest angle of this triangle is 120°, find the value of x. **(3 marks)**

25. Solve the equation $\sqrt{3}\sin\theta - \cos\theta = 0$ for $0 \leq \theta \leq 2\pi$. **(3 marks)**

26. Prove that for any triangle ABC, the following identity holds:

$$a^2 + b^2 + c^2 = 2(bc\cos A + ac\cos B + ab\cos C)$$

(3 marks)

27. Determine the number of solutions to the equation $\sin x = \frac{x}{10}$ for $x \in \mathbb{R}$. **(3 marks)**

28. In a triangle ABC, the sine values of the angles satisfy the ratio $\sin A : \sin B : \sin C = 4 : 5 : 6$. Find the exact value of $\cos A$. **(3 marks)**

29. Show that the equation $\cos\theta - \sin\theta = \sqrt{2}$ has only one solution in the interval $0 \leq \theta \leq 2\pi$, and find this solution. **(3 marks)**

30. In $\triangle ABC$, the side lengths a, b, c form an arithmetic progression with common difference $d > 0$. If the maximum angle of the triangle is 120°, prove that the side lengths must be in the ratio $3 : 5 : 7$. **(3 marks)**

ANSWERS

Section 1

1. Domain: $[-2, 2]$, Range: $[0, 2]$
2. $x = -2$ or $x = 5$
3. 10
4. Domain: $\{x : x \neq 3\}$, Range: $\{y : y \neq 2\}$
5. $x > -2$
6. $f(g(x)) = (x + 3)^2$
7. Odd
8. $[-7, -1]$
9. Vertex: $(2, 0)$, y-intercept: $(0, 2)$
10. $x < -3$ or $x > 3$
11. $x < -2$ or $x > 3$
12. Domain: $\mathbb{R}$
13. Vertex: $(1, -4)$, Intercepts: $(-1, 0), (3, 0), (0, -3)$
14. $-2 \leq x < 3$
15. $f(f(x)) = x$
16. Range: $y \leq 3$
17. Piecewise plot with continuous joint at $(0, -1)$
18. $y = f(-x) + 3$
19. $x = -2, 3$
20. Domain: $x \geq -3$, Range: $y \leq 0$
21. $x < 1$ or $x > 3$
22. Domain: $\{x : x \neq \pm 1\}$, Range: $y < 0$ or $y \geq 1$
23. Shaded region bounded by $y = x^2 - 4$ and $y = |x|$
24. $-4 < k < 8$
25. $f^{-1}(x) = \frac{3x+1}{x-2}$, Domain: $x \neq 2$, Range: $y \neq 3$
26. $-3 \leq x < -2$
27. $x \leq 1$ or $x \geq 3$
28. $f(x) = \frac{x^2-4x+2}{3}$
29. $-6 < m < 2$
30. Discontinuity/hole at $(1, -2)$ and $(1, 2)$, line segments $y = x + 1$ and $y = -x - 1$

Section 2

1. $\frac{5\pi}{4}$

2. $-\frac{\sqrt{3}}{2}$

3. $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

4. 7

5. Amplitude: 3, Period: π

6. $\tan\theta$

7. $\frac{35\sqrt{2}}{2}$ cm^2

8. $-\frac{4}{5}$

9. $\theta = 30°, 150°$

10. $\sin\theta\tan\theta$

11. $\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

12. $2\sec^2\theta$

13. $-\frac{\sqrt{3}}{2}$

14. $B = 49°$ or $131°$

15. $\theta = 90°, 210°, 330°$

16. Cosine wave between $y = -1$ and $y = 3$

17. 12π cm^2

18. $\theta = \frac{7\pi}{24}, \frac{11\pi}{24}$

19. $\sec\theta$

20. 137.4 m

21. $\theta \approx 0.46$ rad, 1.11 rad

22. $2\csc\theta$

23. Wave with amplitude 3, period π, shifted right by $\frac{\pi}{4}$

24. $x = 2.5$

25. $\theta = \frac{\pi}{6}, \frac{7\pi}{6}$

26. Sum of three cosine rules

27. 7

28. $\frac{3}{4}$

29. $\theta = \frac{7\pi}{4}$

30. Side ratio 3 : 5 : 7 from arithmetic progression substitution

FULLY WORKED SOLUTIONS

Section 1

1. For $f(x) = \sqrt{4 - x^2}$, domain requires $4 - x^2 \geq 0 \implies x^2 \leq 4 \implies -2 \leq x \leq 2$. Since $\sqrt{4 - x^2} \geq 0$ and its maximum value occurs at $x = 0$ where $f(0) = 2$, the range is $[0, 2]$.

2. $|2x - 3| = 7 \implies 2x - 3 = 7$ or $2x - 3 = -7$. If $2x - 3 = 7 \implies 2x = 10 \implies x = 5$. If $2x - 3 = -7 \implies 2x = -4 \implies x = -2$.

3. $f(-2) = (-2)^2 - 3(-2) = 4 + 6 = 10$.

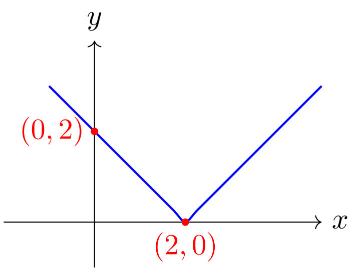
4. For $y = \frac{1}{x-3} + 2$, the denominator cannot be zero, so $x - 3 \neq 0 \implies$ Domain is $x \neq 3$. Since $\frac{1}{x-3} \neq 0$, y cannot equal $2 \implies$ Range is $y \neq 2$.

5. $3 - 2x < 7 \implies -2x < 4 \implies x > -2$.

6. $f(g(x)) = f(x + 3) = (x + 3)^2 = x^2 + 6x + 9$.

7. $f(-x) = (-x)^3 - 3(-x) = -x^3 + 3x = -(x^3 - 3x) = -f(x)$. Since $f(-x) = -f(x)$, the function is odd.

8. $|x + 4| \leq 3 \implies -3 \leq x + 4 \leq 3 \implies -7 \leq x \leq -1$. In interval notation: $[-7, -1]$.

9. 

10. For $f(x) = \frac{1}{\sqrt{x^2-9}}$, we require $x^2 - 9 > 0 \implies (x - 3)(x + 3) > 0 \implies x < -3$ or $x > 3$.

11. $|2x - 1| > 5 \implies 2x - 1 > 5$ or $2x - 1 < -5$. If $2x > 6 \implies x > 3$. If $2x < -4 \implies x < -2$.

12. $g(f(x)) = g(x^2 + 1) = \sqrt{(x^2 + 1) - 1} = \sqrt{x^2} = |x|$. Since $x^2 \geq 0$, the domain of $g(f(x))$ is all real numbers $\mathbb{R}$.

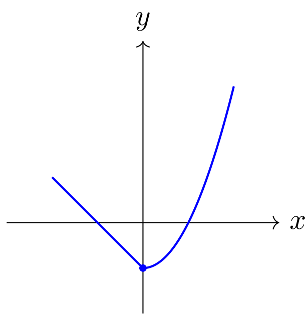
13. $y = (x - 1)^2 - 4$. Vertex is at $(1, -4)$. y-intercept occurs at $x = 0 \implies y = -3$. x-intercepts occur at $y = 0 \implies (x - 1)^2 = 4 \implies x - 1 = \pm 2 \implies x = 3$ or $x = -1$.

14. $\frac{x+2}{x-3} \leq 0$. The critical points are $x = -2$ (where the numerator is 0) and $x = 3$ (where the denominator is 0). Testing the intervals: For $x < -2$, $\frac{-}{-} > 0$. For $-2 \leq x < 3$, $\frac{+}{-} \leq 0$. For $x > 3$, $\frac{+}{+} > 0$. Thus, $-2 \leq x < 3$.

15. $f(f(x)) = f\left(\frac{x}{x-1}\right) = \frac{\frac{x}{x-1}}{\frac{x}{x-1}-1} = \frac{\frac{x}{x-1}}{\frac{x-(x-1)}{x-1}} = \frac{x}{1} = x$.

16. Since $|x+1| \geq 0$ for all real x, $-|x+1| \leq 0 \implies 3 - |x+1| \leq 3$. Thus, the range is $y \leq 3$.

17.



18. Reflection in the y-axis changes $f(x)$ to $f(-x)$. A translation 3 units upwards changes this to $y = f(-x) + 3$.

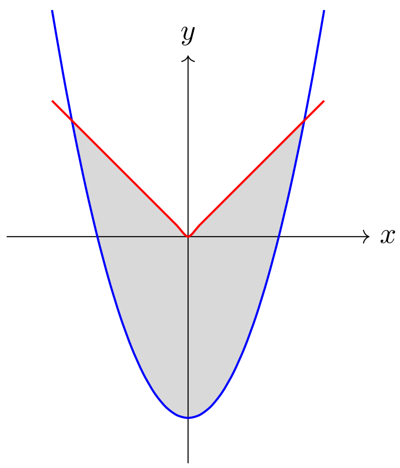
19. $x^2 - 4 = |x+2|$. Case 1: $x \geq -2 \implies x^2 - 4 = x + 2 \implies x^2 - x - 6 = 0 \implies (x-3)(x+2) = 0 \implies x = 3$ or $x = -2$ (both valid). Case 2: $x < -2 \implies x^2 - 4 = -(x+2) \implies x^2 + x - 2 = 0 \implies (x+2)(x-1) = 0 \implies x = -2$ or $x = 1$ (neither is < -2). Solutions: $x = -2, 3$.

20. For $y = -\sqrt{2x+6}$, we need $2x + 6 \geq 0 \implies x \geq -3$. The range is $y \leq 0$ because of the negative sign before the square root.

21. $x^2 - 5x + 6 > |x-3| \implies (x-2)(x-3) > |x-3|$. If $x \geq 3 \implies (x-2)(x-3) > x-3 \implies (x-3)(x-3) > 0 \implies (x-3)^2 > 0 \implies x > 3$. If $x < 3 \implies (x-2)(x-3) > -(x-3) \implies (x-3)(x-1) > 0$. Since $x - 3 < 0$, we must have $x - 1 < 0 \implies x < 1$. Thus, $x < 1$ or $x > 3$.

22. $(f \circ g)(x) = \frac{1}{1-x^2}$. Domain: $1 - x^2 \neq 0 \implies x \neq \pm 1$. Range: Let $y = \frac{1}{1-x^2}$. Since $x^2 \geq 0$ $(x \neq \pm 1)$, $1 - x^2 < 1$. If $0 < 1 - x^2 < 1 \implies y > 1$. If $1 - x^2 < 0 \implies y < 0$. At $x = 0$, $y = 1$. Thus, the range is $y < 0$ or $y \geq 1$.

23.



24. No real roots $\implies \Delta < 0 \implies (k-2)^2 - 4(1)(9) < 0 \implies (k-2)^2 < 36 \implies -6 < k-2 < 6 \implies -4 < k < 8$.

25. Swap x and y: $x = \frac{2y+1}{y-3} \implies x(y-3) = 2y+1 \implies y(x-2) = 3x+1 \implies y = \frac{3x+1}{x-2}$. Rule: $f^{-1}(x) = \frac{3x+1}{x-2}$. Domain of f^{-1} is the range of $f \implies x \neq 2$. Range of f^{-1} is the domain of $f \implies y \neq 3$.

26. $\frac{|x-2|}{(x-2)(x+2)} \geq 1$. Note $x \neq \pm 2$. If $x > 2 \implies \frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2} \geq 1 \implies x+2 \leq 1 \implies x \leq -1$ (no solution since $x > 2$). If $x < 2$ (and $x \neq -2$) $\implies \frac{-(x-2)}{(x-2)(x+2)} = \frac{-1}{x+2} \geq 1$. Since LHS must be positive, $x+2 < 0 \implies x < -2$. Thus $-1 \leq x+2 \implies x \geq -3$. So, $-3 \leq x < -2$.

27. $(f \circ g)(x) = \sqrt{x^2-4x+3}$. Domain requires $x^2-4x+3 \geq 0 \implies (x-1)(x-3) \geq 0 \implies x \leq 1$ or $x \geq 3$.

28. Given (1) $f(x) + 2f(1-x) = x^2$. Substitute $x \leftarrow 1-x$: (2) $f(1-x) + 2f(x) = (1-x)^2 = x^2 - 2x + 1$. Multiply (2) by 2: $2f(1-x) + 4f(x) = 2x^2 - 4x + 2$. Subtract (1) from this: $3f(x) = x^2 - 4x + 2 \implies f(x) = \frac{x^2-4x+2}{3}$.

29. $x^2 - 2x - 1 = mx - 5 \implies x^2 - (2+m)x + 4 = 0$. No intersection $\implies \Delta < 0 \implies (2+m)^2 - 16 < 0 \implies (2+m)^2 < 16 \implies -4 < 2+m < 4 \implies -6 < m < 2$.

30. $y = \frac{|x^2-1|}{x-1}$. Note $x \neq 1$. If $x > 1$ or $x \leq -1$, $y = \frac{x^2-1}{x-1} = x+1$. If $-1 < x < 1$, $y = \frac{-(x^2-1)}{x-1} = -(x+1) = -x-1$. There are holes at $(1,2)$ and $(1,-2)$. Intercepts at $(0,-1)$ and $(-1,0)$.

Section 2

1. $225° \times \frac{\pi}{180°} = \frac{5\pi}{4}$ radians.

2. $\sin\left(\frac{5\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$ (since $\frac{5\pi}{3}$ is in the 4th quadrant).

3. $\cos\theta = -\frac{1}{2}$. The reference angle is $\frac{\pi}{3}$. In quadrants 2 and 3: $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ and $\theta = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$.

4. By the Cosine Rule: $x^2 = 5^2 + 8^2 - 2(5)(8)\cos(60°) = 25 + 64 - 80(0.5) = 89 - 40 = 49 \implies x = 7$.

5. For $y = 3\sin(2x)$, Amplitude $= 3$. Period $= \frac{2\pi}{2} = \pi$.

6. $\frac{\sin(\pi-\theta)}{\sin(\frac{\pi}{2}-\theta)} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

7. Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(7)(10)\sin(45°) = 35 \times \frac{\sqrt{2}}{2} = \frac{35\sqrt{2}}{2}$ cm^2.

8. Since $\frac{\pi}{2} < \theta < \pi$, θ is in the 2nd quadrant where cosine is negative. Using $\tan\theta = -\frac{3}{4}$, the hypotenuse is 5. Hence, $\cos\theta = -\frac{4}{5}$.

9. $2\sin\theta = 1 \implies \sin\theta = \frac{1}{2} \implies \theta = 30°, 150°$.

10. $\cos(90° - \theta)\tan(\theta) = \sin\theta\left(\frac{\sin\theta}{\cos\theta}\right) = \sin\theta\tan\theta$.

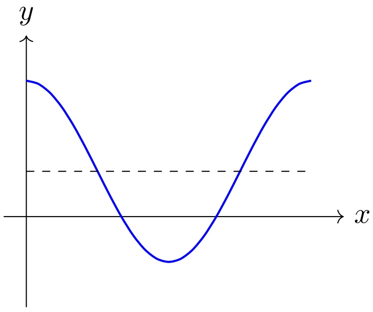
11. $\tan^2\theta = 3 \implies \tan\theta = \pm\sqrt{3}$. Reference angle is $\frac{\pi}{3}$. In all four quadrants: $\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

12. LHS $= \frac{(1+\sin\theta)+(1-\sin\theta)}{(1-\sin\theta)(1+\sin\theta)} = \frac{2}{1-\sin^2\theta} = \frac{2}{\cos^2\theta} = 2\sec^2\theta =$ RHS.

13. $\cos\left(-\frac{7\pi}{6}\right) = \cos\left(\frac{5\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.

14. By the Sine Rule: $\frac{\sin B}{12} = \frac{\sin 30°}{8} \implies \sin B = \frac{12\times 0.5}{8} = 0.75$. $B_1 \approx 49°$ or $B_2 = 180° - 49° = 131°$. Both are valid since $A + B < 180°$.

15. Substitute $\cos^2\theta = 1 - \sin^2\theta \implies 2(1-\sin^2\theta) + \sin\theta - 1 = 0 \implies 2\sin^2\theta - \sin\theta - 1 = 0 \implies (2\sin\theta + 1)(\sin\theta - 1) = 0$. If $\sin\theta = 1 \implies \theta = 90°$. If $\sin\theta = -0.5 \implies \theta = 210°, 330°$.

16. 

17. $l = r\theta \implies 4\pi = 6\theta \implies \theta = \frac{2\pi}{3}$ radians. Area $= \frac{1}{2}r^2\theta = \frac{1}{2}(36)\left(\frac{2\pi}{3}\right) = 12\pi$ cm^2.

18. $\sin\left(2\theta - \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}$. Since $0 \le \theta \le \pi$, we have $-\frac{\pi}{4} \le 2\theta - \frac{\pi}{4} \le \frac{7\pi}{4}$. Thus, $2\theta - \frac{\pi}{4} = \frac{\pi}{3} \implies 2\theta = \frac{7\pi}{12} \implies \theta = \frac{7\pi}{24}$. Or $2\theta - \frac{\pi}{4} = \frac{2\pi}{3} \implies 2\theta = \frac{11\pi}{12} \implies \theta = \frac{11\pi}{24}$.

19. $\cos\theta + \sin\theta\left(\frac{\sin\theta}{\cos\theta}\right) = \frac{\cos^2\theta+\sin^2\theta}{\cos\theta} = \frac{1}{\cos\theta} = \sec\theta$.

20. Let d be the horizontal distance. $\tan(20°) = \frac{50}{d} \implies d = \frac{50}{\tan(20°)} \approx 137.4$ m.

21. Divide by $\cos^2\theta$: $2\tan^2\theta - 5\tan\theta + 2 = 0 \implies (2\tan\theta - 1)(\tan\theta - 2) = 0$. If $\tan\theta = 0.5 \implies \theta \approx 0.46$ rad. If $\tan\theta = 2 \implies \theta \approx 1.11$ rad.

22. LHS $= \frac{\sin^2\theta+(1+\cos\theta)^2}{\sin\theta(1+\cos\theta)} = \frac{\sin^2\theta+1+2\cos\theta+\cos^2\theta}{\sin\theta(1+\cos\theta)} = \frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)} = \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} = 2\csc\theta =$ RHS.

23. $y = 3\sin\left(2(x - \frac{\pi}{4})\right)$. Amplitude $= 3$, Period $= \pi$, Phase shift $= \frac{\pi}{4}$ to the right.

24. The largest angle is $120°$, opposite the largest side $a = x+1$. By the Cosine Rule: $(x+1)^2 = x^2+(x-1)^2-2x(x-1)\cos(120°)$ $x^2+2x+1 = x^2+x^2-2x+1+x(x-1) \implies 2x^2-5x=0 \implies x=2.5$ (since $x>1$ for positive sides).

25. $\sqrt{3}\sin\theta = \cos\theta \implies \tan\theta = \frac{1}{\sqrt{3}}$. In quadrants 1 and 3: $\theta = \frac{\pi}{6}, \frac{7\pi}{6}$.

26. By the Cosine Rule: $bc\cos A = \frac{b^2+c^2-a^2}{2}$, $ac\cos B = \frac{a^2+c^2-b^2}{2}$, $ab\cos C = \frac{a^2+b^2-c^2}{2}$. Summing these yields: $bc\cos A + ac\cos B + ab\cos C = \frac{a^2+b^2+c^2}{2} \implies 2(bc\cos A + ac\cos B + ab\cos C) = a^2+b^2+c^2$.

27. The solutions correspond to intersections of $y=\sin x$ and $y=\frac{x}{10}$. Since $|\sin x| \leq 1$, intersections must occur in $[-10, 10]$. There is 1 solution at $x=0$. For $x>0$, the line crosses the wave once in $[0,\pi]$, and twice in $[2\pi, 3\pi]$ (since $3\pi \approx 9.42 < 10$). Total of 3 positive solutions. By symmetry, there are 3 negative solutions. Total $= 1+3+3 = 7$ solutions.

28. By the Sine Rule, $a:b:c = \sin A : \sin B : \sin C = 4:5:6$. Let $a=4k, b=5k, c=6k$. $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{25k^2+36k^2-16k^2}{2(5k)(6k)} = \frac{45}{60} = \frac{3}{4}$.

29. Divide by $\sqrt{2}$: $\frac{1}{\sqrt{2}}\cos\theta - \frac{1}{\sqrt{2}}\sin\theta = 1 \implies \cos\left(\theta+\frac{\pi}{4}\right) = 1$. For $0 \leq \theta \leq 2\pi \implies \frac{\pi}{4} \leq \theta + \frac{\pi}{4} \leq \frac{9\pi}{4}$. The only solution is $\theta + \frac{\pi}{4} = 2\pi \implies \theta = \frac{7\pi}{4}$.

30. Let the side lengths be $x-d, x, x+d$. The largest angle is $C=120°$, opposite to $c=x+d$. Cosine Rule: $(x+d)^2 = (x-d)^2+x^2-2x(x-d)\cos(120°) \implies x^2+2xd+d^2 = x^2-2xd+d^2+x^2+x(x-d) \implies 2x^2-5xd=0 \implies x=2.5d$. Hence, side lengths are $1.5d, 2.5d, 3.5d \implies$ ratio is $3:5:7$.